

Symbolic Compression (Structural Edition)

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Abstract

Symbolic Compression formalises how structural information is reduced into canonical symbolic form. It defines compression gradients, canonical symbol formation, motif-to-symbol mapping, collapse-compression interaction, and horizon-bounded symbolic behaviour. This edition presents the basis-independent formulation suitable for academic publication and cross-domain application.

1 Introduction

Symbolic Compression describes how structural information becomes stable symbolic representation. It governs:

- Compression Gradient
- Canonical Symbol Formation
- Motif-to-Symbol Mapping
- Collapse-Compression Interaction
- Horizon-Bounded Symbolic Behaviour
- Symbol Drift
- Symbol Interference

2 Compression Gradient

Let I denote information density and C compression capacity. Compression gradient:

$$G = I - C.$$

Compression occurs when:

$$G > 0.$$

Compression radius:

$$R = f(G).$$

3 Canonical Symbol Formation

Symbol formation condition:

$$S > S_{\min}.$$

Symbol radius:

$$K = f(S).$$

4 Motif-to-Symbol Mapping

Mapping function:

$$\Sigma = h(M).$$

Mapping stability:

$$\frac{d\Sigma}{dt} > 0.$$

5 Collapse-Compression Interaction

Collapse-compression coupling:

$$\Gamma = g(V_c, G).$$

6 Horizon-Bounded Symbolic Behaviour

Symbolic opacity:

$$O = 1 - \frac{S_{\text{out}}}{S_{\text{in}}}.$$

7 Symbol Drift

Drift velocity:

$$V_d = \frac{dK}{dt}.$$

8 Symbol Interference

Interference strength:

$$I = h(\Sigma_1, \Sigma_2, d).$$

9 Conclusion

Symbolic Compression defines compression gradients, canonical symbol formation, motif-to-symbol mapping, collapse-compression interaction, and horizon-bounded symbolic behaviour. This edition presents the basis-independent formulation suitable for academic publication and cross-domain application.